used by the majority of the members. Width of bridge in the back tube sheet varies from 11/16 inch to 15/16 inch. Three-fourths inch is used by the majority of the members.

Flue Setting—Two members advise that they have tried out soft iron ferrules with very little success. One member advises that they had a limited experience trying to set flues in

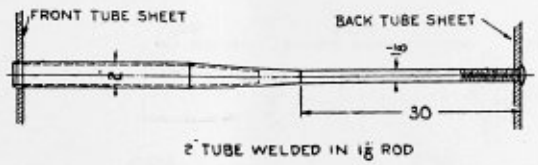


FIG. 20

the back flue sheet without any coppers, but found that it did not work satisfactorily, and that they are now experimenting with a combined copper and iron ferrule, but have not had these under a test long enough to make any report. One member advises that they use soft iron shims on the front flues to avoid excessive expanding.

Setting of Flues in the Front Tube Sheet—The consensus

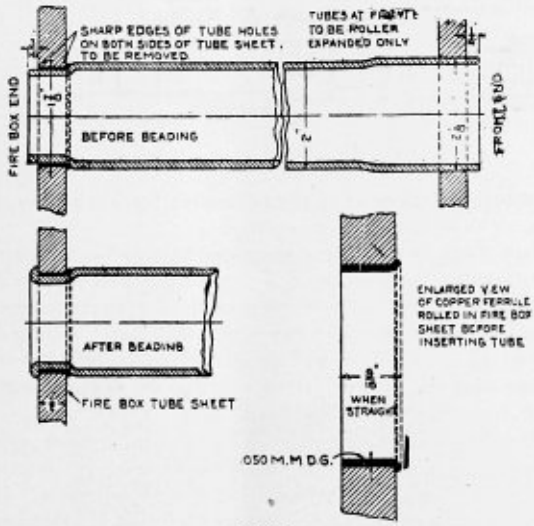


FIG. 21

of opinion of the roads reporting is that tubes should be rolled in the front tube sheet without any ferrules, and 10 percent of the flues, equally distributed, should be beaded with a beading tool.

Tube Setting in Back Sheet—Fig. 21 shows the practice followed out by some of the members in setting the tubes, which are giving good results. Fig. 24 shows the application, maintenance and removal of flues in locomotive boilers as used by three members. Tube setting of this kind gives a large bridge, good circulation of water at the back tube sheet, and at the same time the inside of the flue is reduced at the end, so that any cinders which will pass through the swedged end will not clog the flues. The instructions in connection with Fig. 24 are as follows:

Holes in the new firebox flue sheet shall be 1⅞ inches diameter. Holes in the new smokebox flue sheet shall be 2 1/32 inches diameter. Holes in an old flue sheet more than 1/32 inch out of round must be reamed (reamer Fig. 1). Inside and outside edges of the flue holes in

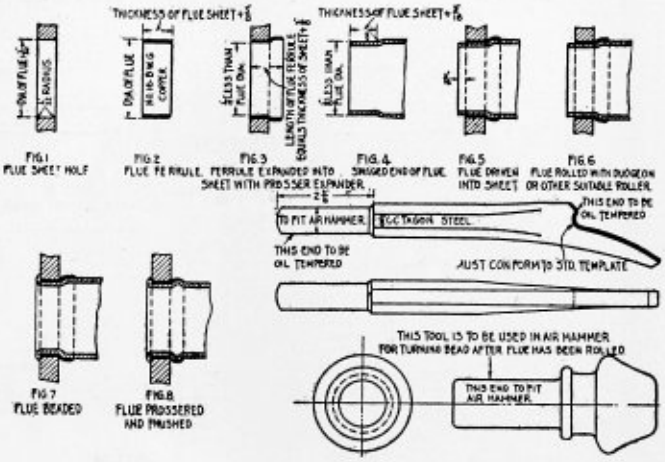


FIG. 23

both sheets must be slightly rounded to remove any sharp edges. The following is the method used:

Flue-sheet Holes (Fig. 1)—Care should be taken that holes in flue sheet are true, smooth and free from burrs and sharp edges. It is desirable that the flue hole have a fillet especially on the water side of about 1/32 inch radius. A sharp edge around the hole often cuts the ferrule in two, even cutting into the flue. The diameter of the flue hole should be the same as outside diameter of flue plus 1/32 inch.

Copper Ferrule (Fig. 2)—For New York this should not be far different from No. 16 B. W. G. In old flue sheets when the flue holes are large it is desirable to use enough heavier ferrules to bring the internal diameter of ferrule when expanded into flue sheet to ⅛ inch less than flue diameter. The ferrule should be set into flue hole flush with fireside and expanded into place with Prosser expander, Fig. 3.

Swaging Flue (Fig. 4)—Great care should be taken to give the swage the right length and to have it terminate in as abrupt a shoulder as possible. The flue, after it is swaged, should be annealed and the scale removed from the portion entering the sheet. Grinding this scale off by machine is desirable, but in the absence of a grinder the removal of the scale with a file will answer. The flue when driven into the sheet (Fig. 5) should extend 3/16 inch through the sheet on the fire side.

Rolling (Fig. 6)—The front end of the flues having been shimmed, when necessary, both the back and front ends are rolled with the dudgeon or other suitable roller, with proper appliances they may be safely and economically rolled by air; the rolling of the back ends must be done by an experienced boiler maker, and the motor must not be larger than a Thor No. 22, or its equivalent.

Beading (Fig. 7)—This may now be done with a beading tool in an air hammer. Do as little hammering with the beading tool as possible, and do not work bead to a feather edge, but leave as much metal as possible for future calking.

Expanding (Fig. 8)—Particular stress should be placed upon this, the most important step in the flue-setting process. Care should be taken that the flue shoulder, when heavily expanded, bears snugly and firmly against the flue sheet. Hammer not to exceed 4 pounds after prossering the front ends, then bead a sufficient number for safety which finishes the flue setting. Back ends of flues or safe ends when used in back flue sheet must be cut off square in pipe machine and burr removed by reamer. Wheel-flue cutters must not be used on account of leaving a heavy burr which splits when beading over

NOTE.—Air motor and roller are only to be used on initial setting.

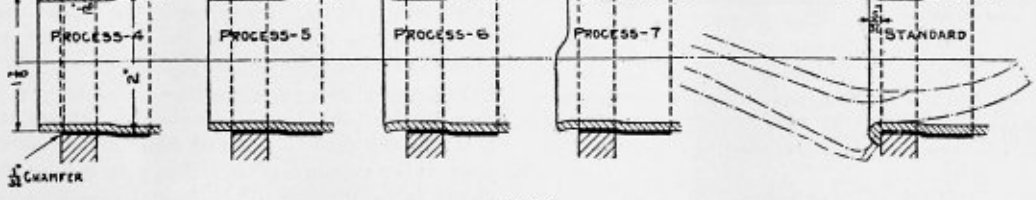


FIG. 22

In reworking flues, the Prosser expander only is to be used, and rollers must not be used, as they force the bead away from the sheet.

Copper ferrules 1¾ inches inside diameter, ½ inch long, .095 inch in thickness shall be used in firebox flue sheet only. Ferrules shall be secured in place with straight expander, Fig. 2, taking care that shoulder of expander is tight against flue sheet, which shall bring edge of ferrule 1/32 inch from fire side of sheet.

Flue safe ends should be 5 inches in length, except that new iron flues shall have safe ends 4 inches long, and when these flues are first removed the 4-inch ends will be cut off completely and 6-inch ends applied, after which 5-inch ends should be used. Flues shall be swaged as

tubes. Some engineer will say that you can't stall them; that is, they pull more. Starting a train is exactly the same as it is with a saturated steam engine, because if you are carrying 200 pounds of steam you are only getting 200 pounds pressure on each square inch of your piston, but the advantage of superheating steam is that it eliminates the saturation and the steam works expansively, and you don't need so much steam as when you are using saturated steam, which they brought out in their report very nicely.

Now, on a road which I was formerly connected with, an engineer had a device whereby the water was put into the engine at boiling heat; in fact, it got up sometimes as high as 225 degrees of heat going into the boiler. The system consisted of exhaust steam passing through a pipe alongside of the engine and back to the tank and going around coils of pipe, and inside the pipe was the water, which would be pumped into boiler and got the heat from the exhaust steam. Tests on that system of superheating showed a saving of fuel and also a saving of flues, and on one engine to which it was applied there was hardly any work for a couple of years. I don't think the flues were expanded but a few times, while on other types of engines running in the same district the flues had to be expanded quite frequently, which shows that the injecting or putting water into the boiler over the boiling point is a considerable saving in fuel and in the up-keep of a locomotive.

Mr. G. E. Ryder, a representative of the Locomotive Superheater Company, New York, gave a detailed description of the type of superheater which this company manufactures, together with its advantages, as compared with other types. A paper covering in substance Mr. Ryder's remarks is given in full on page 178.

The Chairman: The standard practice, according to Mr. Lowe's report, is that they apply these superheater tubes without copper ferrules; have they applied any with copper ferrules so as to note the difference?

Mr. Lowe: We have not used copper, although it has been tried on one engine, due to an error or having stock of a size which was smaller, and we made that up with the ferrule, but we find that we don't need the copper ferrule; we find that we get good results whether the water is good or bad. There are some cases where we would not have to calk the flue in a year or two years, and in other cases we might have to give them attention now again after they had been in service for some time, but we are practically free from trouble with reference to leaking in service.

Mr. Ryder: I know of some engines in the States that are running without copper ferrules and have not given trouble, and I know of some that are running with copper ferrules and are giving trouble; but the bulk of our experience points toward the use of copper ferrules and a little heavier copper ferrule than is used with the small flues.

The use of superheated steam is responsible for a decrease in the boiler maintenance cost; it makes it possible to reduce the boiler pressure without affecting the smartness of the engine, because larger cylinders can be used, since the condensing surface requires no consideration. A further reduction in boiler cost is obtained because the flue maintenance cost is lower in severe conditions. There is one more point; the superheater engine will develop the same power on 35 percent less water, and you should be able to lengthen the periods or increase the mileage between washouts if your engines are developing the same power in proportion to the water you save, and that has been the experience in some cases that I know of.

Mr. Gray: We have engines with 29-inch cylinders and 62-inch drivers, carrying 160 pounds of steam pressure. I guess that is as low as anyone using superheated engines. There is one thing I have noticed about them; once in a while one of those engines will come in with all the large flues leaking, or possibly two rows of them, sometimes the whole set, and no small ones leaking. In every case we have tested the engine, and found the superheater practically leaking at one end, and we have never found any of those large flues leaking unless there was a leak in the front end. Sometimes there will be three or four or five simmering a little bit when the small flues are leaking, but when we found the large flues leaking and no small ones, or only a few, we've always got a leak in the front end.

Mr. Ewell: He didn't say whether the fire-box end was the large or small end.

Mr. Ryder: The small end. We have 24 inches between the flue sheet and the back end of the superheated coil or unit, as we call it; we have 18 inches from the small portion of the flue to the end of the superheater unit, so you can apply about 18 inches of safe-end before you are in any danger of coming in contact with the unit.

A Member: What end would you suggest to safe-end?

Mr. Ryder: The small end, the fire-box end, because the material in the fire-box end is subjected to the most severe part of the flue and you are supplying new material where the conditions are severest. Another reason is that the small end of the tube is smaller in diameter, of course, and the cost of safe-ending material is less. The third reason is that the 4½-inch diameter is standard, and it is only necessary to carry one size of safe-end for both 5⅜ and 5½-inch flues.

Mr. Lowe: We are not following the practice on Western lines of welding the safe-end onto the 5-inch flues at the fire-box end. We have reversed our practice in the case of the 5-inch flues, and we have been using those flues since about 1905, and we have had no bad results whatever from welding on to the smoke-box end. By doing so with the machinery that we have we save one operation.

Mr. Goodwin: Do you get any more mileage out of a superheated tube than out of your small tube?

Mr. Lowe: Yes.

Mr. Goodwin: You don't have to remove the superheated tube every time?

Mr. Lowe: Not every time.

Mr. Goodwin: Do you get two settings out of your superheated tube?

Mr. Lowe: Frequently. Mr. Ryder mentioned about different sized beading tools. We have two size beading tools only: that is, a No. 1 and a No. 2. We used to carry the different sized beading tools, and we came to this conclusion, that if it was right to put in a tube with a standard sized beading tool that was the size to maintain it under, and when it grew to be too big we made it the right size for the tool. Therefore we carry a beading tool to suit the 2 and 2¼-tube and another to suit the larger tube, which necessarily is a much larger tool.

Mr. Hennessy: How do you save the company money by welding on to the front end and using the back end of the flue by heating and upsetting? Does that retain the life of that material by heating and upsetting? Suppose that flue is a little thin, you are supposed to have a No. 7 gage in there. That flue is reduced down to No. 12, No. 13 or No. 14 gage; how are you going to upset that back far enough in there to retain the strength of that flue? If such is the case, saving the company money, why do you buy this new material for safe-ending? Why don't you use your old tube, if you can handle that that way? By not using copper ferrules, Mr. Lowe says they run their flues from a year to two years without leakage. That is something remarkable. We can't run our flues that many weeks without leakage, and we use copper ferrules, 13 and 15-wire gage. I believe, in welding and safe-ending flues, the proper place is in the back end to put your new material.